

Robust 3D Human Pose Estimation from mmWave Radar via Spatio-Temporal Representation Learning

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The Bottleneck of Current mmWave HPE

- **Point Cloud Sparsity & Noise:** mmWave point clouds are **sparse, noisy**, and prone to multipath interference and missing detections.
- **Temporal & Domain Instability:** **Temporal gaps** and **cross-domain distribution shifts** (subjects/environments) severely degrade generalization.
- **Tight-Coupling Drawbacks:** Tightly coupling spatial and temporal features causes **error accumulation** and unstable motion under fragmented radar inputs.

TL;DR

We propose a **stage-wise** spatio-temporal representation learning framework that explicitly decouples **spatial structure learning**, **temporal motion consistency modeling**, and **spatio-temporal integration**, followed by a **parameter-efficient conditional diffusion module** for pose refinement. On the mmFi benchmark, our method achieves **up to a 22.5% reduction** in MPJPE under the challenging cross-environment setting, demonstrating superior robustness and generalization.

How Our Framework Works

Data Preprocessing & Normalization

Applies **sliding-window** aggregation to stabilize local temporal context and **standardizes point counts** with 5 channels (x, y, z, v, I).

MMCL

Self-supervised temporal MAE reconstructing masked frame features via a PreNorm Transformer to enforce **motion coherence** under missing radar inputs.

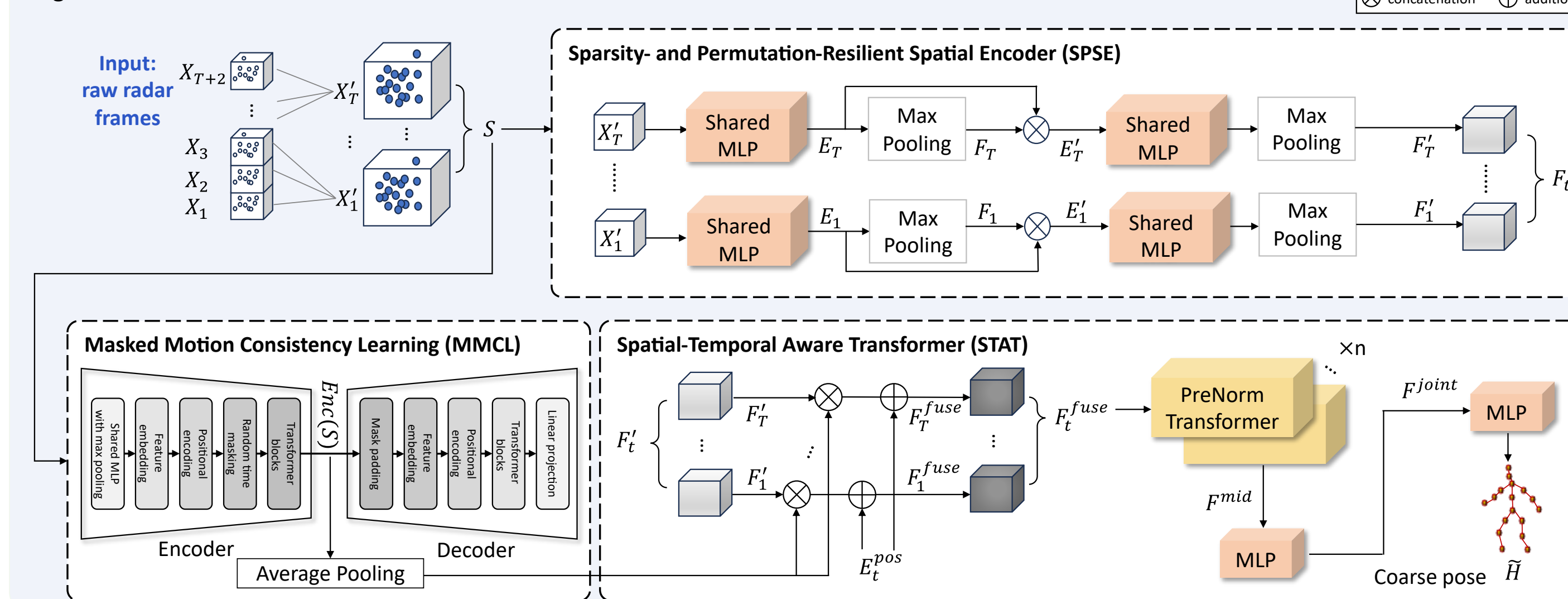
TJA

Captures **joint-wise motion continuity** over past time steps to suppress frame-wise jitter.

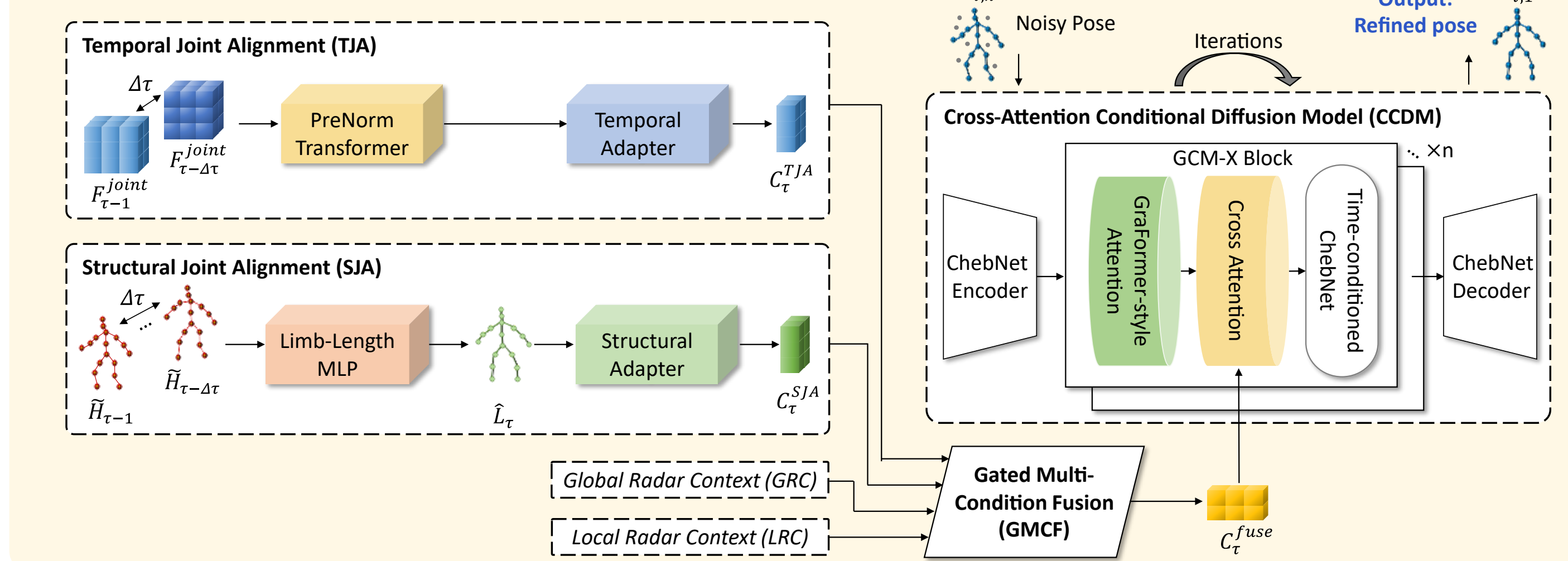
SJA

Estimates **invariant skeletal limb lengths** via Limb-Length MLP and projects them to joint nodes via an incidence matrix, ensuring anatomical plausibility.

Stage 1: mmWave Feature Extraction



Stage 2: Diffusion-Based Human Pose Refinement



Progressive 3-Phase Training Strategy

- **Phase 1 (Self-Supervised):** Train MMCL via feature reconstruction loss and uniformity regularization.
- **Phase 2 (Supervised Stage 1):** Freeze MMCL; train SPSE, STAT, and regression head with L2 pose loss.
- **Phase 3 (Refinement Training):** Freeze Stage 1; train CCDM conditional diffusion module using noise prediction objective.

SPSE

Employs shared point-wise MLPs and max pooling to aggregate global-local representations, ensuring **permutation invariance** and **structural stability** under sparse point clouds.

STAT

Integrates **spatial descriptors** and **global motion context** to regress an initial coarse 3D pose.

CCDM

Combines ChebNet graph convolutions, self/cross-attention GCM-X blocks, and Chebyshev decoders to iteratively **refine the noisy pose** into a clean, smooth 3D skeleton.

GMCF

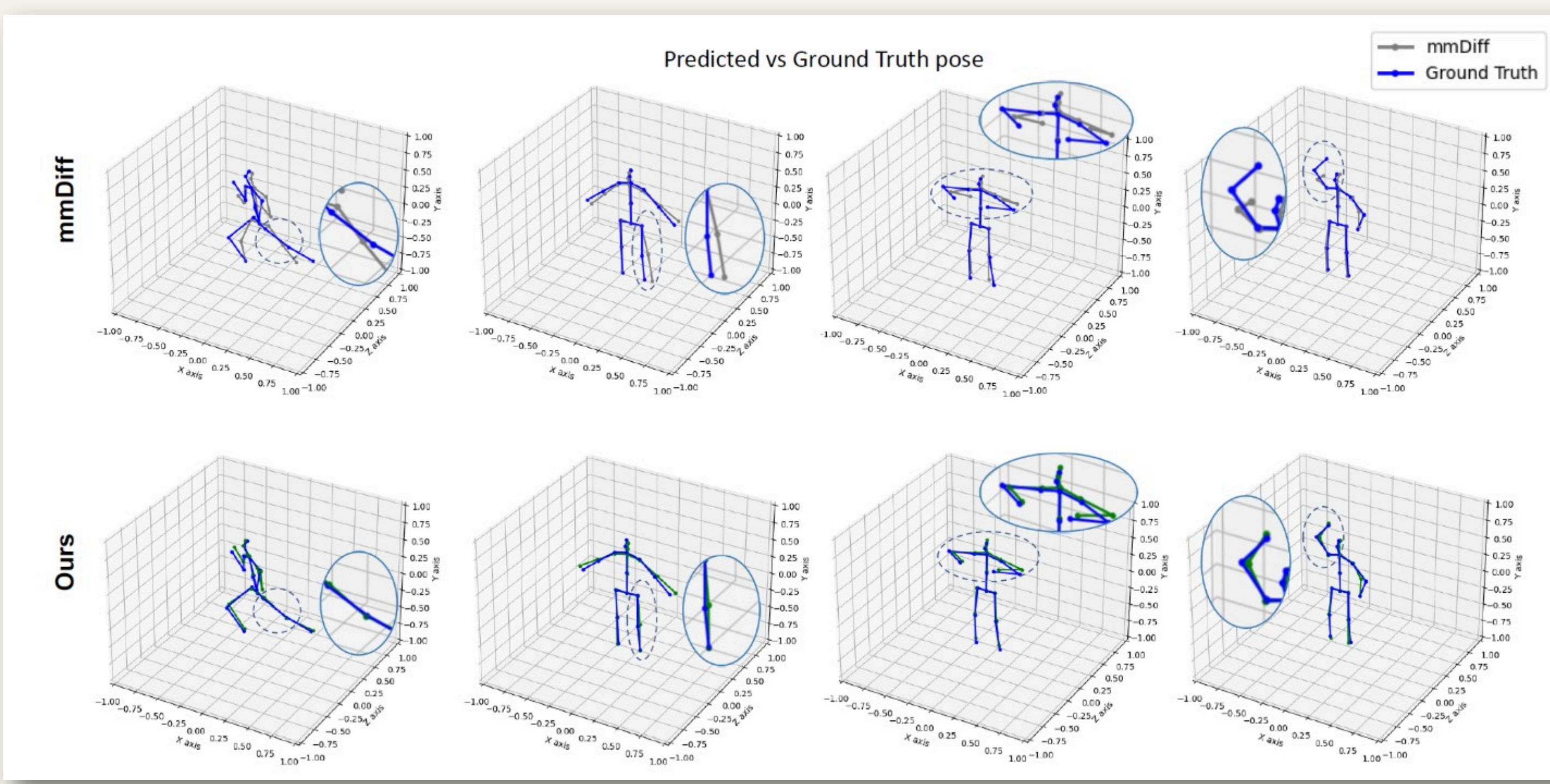
Fuses **heterogeneous conditioning tokens** for flexible cross-attention injection.

State-of-the-Art Comparison on mmFi Benchmark

Our framework achieves a **22.5% MPJPE error reduction** under the hardest cross-environment setting (S3/P3), outperforming both PointTransformer and mmDiff baselines.

Method	Setting	Protocol 1		Protocol 2		Protocol 3	
		MPJPE	PA-MPJPE	MPJPE	PA-MPJPE	MPJPE	PA-MPJPE
PointTransformer [22]	S1	73.09±2.70*	55.60±1.40*	73.38±0.99	57.23±0.74	71.75±0.67	56.08±0.59
	S2	75.96±6.90*	58.70±4.30*	71.30±0.60	55.22±0.64	72.76±1.27	57.59±1.13
	S3	88.28±4.50*	68.79±2.79*	97.48±3.68	78.81±3.42	96.59±2.57	77.27±2.29
mmDiff [5]	S1	65.26±0.11*	50.35±0.09*	66.91±0.16	51.92±0.96	65.00±0.18	50.76±0.12
	S2	65.62±0.24*	50.23±0.24*	63.95±0.58	49.70±0.18	66.91±0.16	51.91±0.96
	S3	82.73±0.62*	63.87±0.26*	91.29±1.53	73.18±0.70	93.35±2.17	74.74±2.02
Ours	S1	61.62±0.17	46.21±0.11	57.54±0.07	43.25±0.02	57.19±0.05	42.98±0.06
	S2	59.64±0.07	44.56±0.05	54.87±0.12	41.79±0.34	56.24±0.05	42.41±0.05
	S3	71.66±0.08	52.01±0.03	72.67±0.17	55.40±0.10	72.07±0.06	53.55±0.07

Under challenging cross-scene domain shifts, our framework maintains **high structural coherence** and **joint alignment**, whereas mmDiff suffers from severe limb distortion and joint misalignment (circled).



Why Our Method Works?

	Setting 1		Setting 2		Setting 3	
	MPJPE	PA-MPJPE	MPJPE	PA-MPJPE	MPJPE	PA-MPJPE
(a) Effects of the input sequence length T						
$T = 15$	57.90±0.23	43.74±0.06	56.33±0.17	42.59±0.08	73.89±0.34	55.84±0.15
$T = 25$	57.60±0.09	43.39±0.03	56.64±0.08	42.78±0.05	72.52±0.12	54.09±0.12
$T = 35$	57.93±0.07	43.67±0.05	57.50±0.21	43.54±0.16	73.72±0.21	55.32±0.17
(b) Effects of individual components						
w/o SPSE	117.14±0.17	103.08±0.37	116.19±0.50	105.04±0.97	118.71±0.12	102.3±0.10
w/o STAT	67.13±0.16	51.98±0.10	66.35±0.33	52.36±0.16	90.24±0.13	71.85±0.12
w/o MMCL	57.59±0.07	43.05±0.19	56.82±0.21	42.83±0.10	73.16±0.46	54.91±0.55
w/o Diff.	59.52±0.15	44.55±0.09	58.62±0.17	44.00±0.11	73.77±0.21	55.17±0.35
Full model	57.19±0.05	42.98±0.06	56.24±0.05	42.41±0.05	72.07±0.06	53.55±0.07

Optimal Temporal Sequence Length T

$T = 25$ provides the **best trade-off** between temporal motion context and noise suppression (outperforming $T = 15$ and $T = 35$).

Impact of Core Architectural Components

- **w/o SPSE (Severe Drop):** S1 MPJPE surges to **117.14mm** (+104.8%) – proves spatial encoding is essential for sparse radar inputs.
- **w/o STAT (Cross-Domain Drop):** S3 MPJPE degrades to **90.24mm** – highlights the need for explicit spatio-temporal fusion.
- **w/o MMCL (Generalization Drop):** Performance degrades across all splits – validates masked feature reconstruction for temporal motion consistency.
- **w/o Diffusion (Fine Jitter Drop):** S3 MPJPE increases from **72.07mm** → **73.77mm** – confirms TJA & SJA successfully rectify joint jitter and anatomical implausibility.

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ICPR Paper:



RRPR Companion Paper:



Code:

